What's Wrong With Science?

Session 2: Who should pay for science? The economics of science funding

Professor David Gresham
Fall 2026

Tonight's Guest



Nate Williams

Communications Director
Alfred P. Sloan Foundation

You are Congress.

The federal government will spend about $7.4 trillion this year. We are going to shrink that to $100.



1. Form a group with everyone in your row. Move to the north if you are are in odd row, south if even row.
2. Divide $100 across the 8 spending categories on the next slide. Whole dollars only; it must add up to exactly $100.
3. Choose one person to be the scribe and reporter. Write down the one-sentence reason for your science number.
4. Submit your science budget on Poll Everywhere. You have 8 minutes to discuss.

Divide your $100 across these 8 categories

Social Security

Retirement and disability checks to ~70 million people

$ ___

Health care

Medicare, Medicaid, ACA subsidies, CHIP

$ ___

National defense

Military pay, operations, weapons, bases

$ ___

Interest on the debt

Payments to holders of U.S. Treasury bonds

$ ___

Veterans & retirees

VA health care and benefits; civilian and military pensions

$ ___

Income security

SNAP (food stamps), unemployment insurance, housing aid, child tax credit

$ ___

Science & research

NIH, NSF, NASA, DOE labs, defense R&D, university grants

$ ___

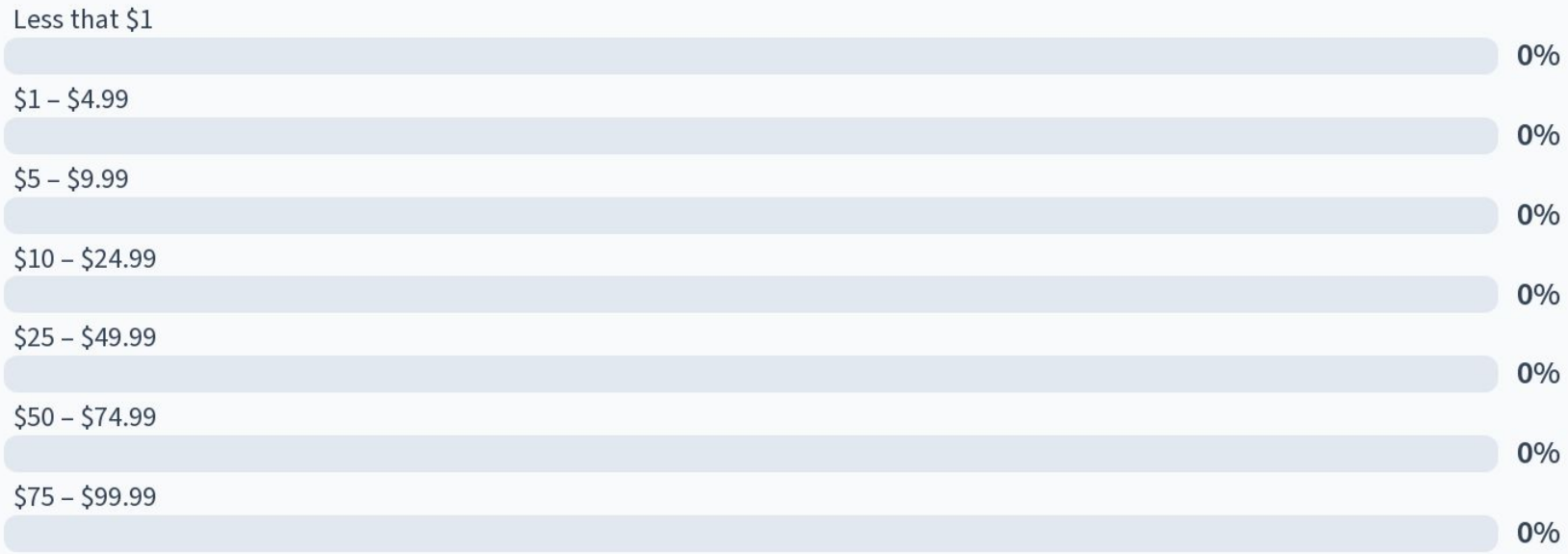
Everything else

Education, transportation, agriculture, foreign aid, EPA, FBI, courts, national parks…

$ ___

Rules: whole dollars, must total exactly $100

How much of the US Federal Budget ($100 total) should be spent on Science and Research?



Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Justify your decision:

- Group allocating the highest amount
- Group allocating the lowest amount

You are a grant agency

You have $10 million to spend



1. Starting from the front row form a group with the row behind you.
2. Select which grants to fund (you can't go over $10 million, but you can be under)
3. Choose one person to be the reporter of the grants you will award.
4. Submit your science budget on Poll Everywhere. You have 8 minutes to discuss.

What will you fund? (Budget = $10 million)

(1) Basic yeast cell-division research ($2M)

(2) Phase I pediatric leukemia drug trial ($4M)

(3) Malaria vaccine work for sub-Saharan Africa ($3M)

(4) Women's heart-disease diagnostic study ($2M)

(5) mRNA-vaccine for cancer immunotherapy from unknown junior scientist ($3M)

(6) Replication project re-testing 50 influential psychology findings ($1M)

(7) AI drug-discovery startup partnership ($4M)

(8) Community air-quality study in the Bronx ($1M)

What will you fund? (Budget = $10 million)

Basic yeast cell-division research ($2M) 0%

Phase I pediatric leukemia drug trial ($4M) 0%

Malaria vaccine work for sub-Saharan Africa ($3M) 0%

Women's heart-disease diagnostic study ($2M) 0%

mRNA-vaccine for cancer immunotherapy from unknown junior scientist ($3M) 0%

Replication project re-testing 50 influential psychology findings ($1M) 0%

AI drug-discovery startup partnership ($4M) 0%

Community air-quality study in the Bronx ($1M) 0%

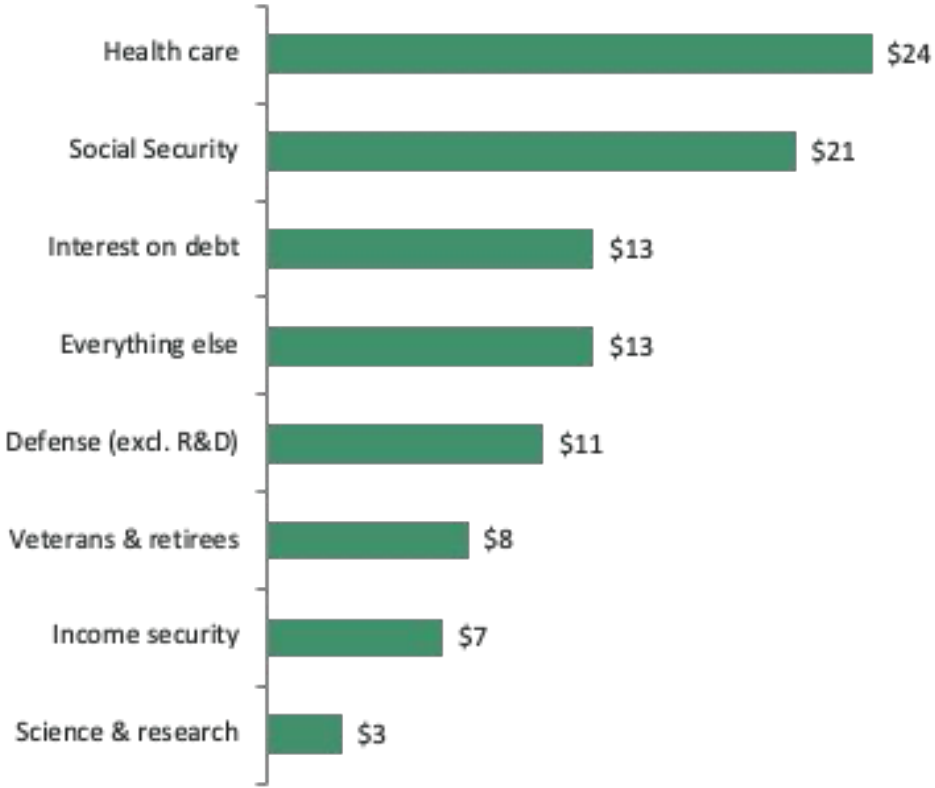
Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

What are the trends?

- Any grants that everyone agrees should be funded? Why is this?

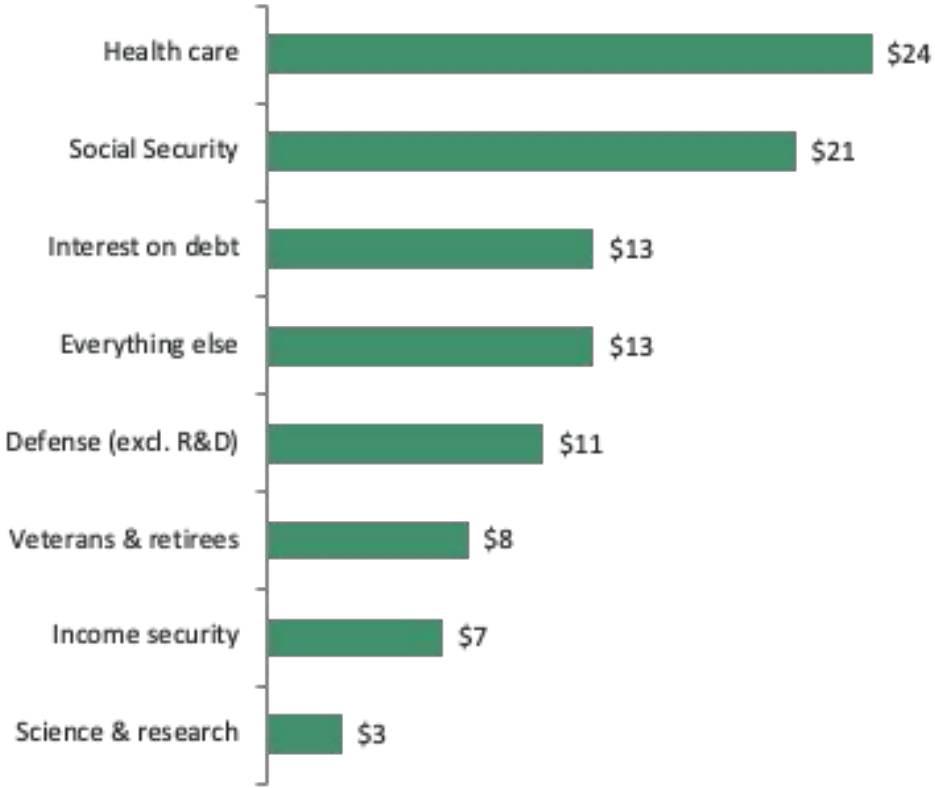
- Which Grant was the least popular? Why?

What Congress actually does with $100



Sources: CBO Budget & Economic Outlook 2026–2036; CBPP; CRS R48694 (FY2026 R&D). Rounded.

What Congress actually does with $100



Science & research

$2.70

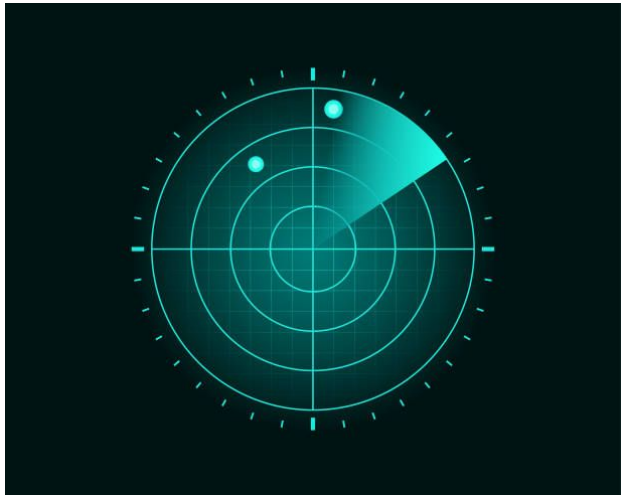
of every $100 — about $200 billion

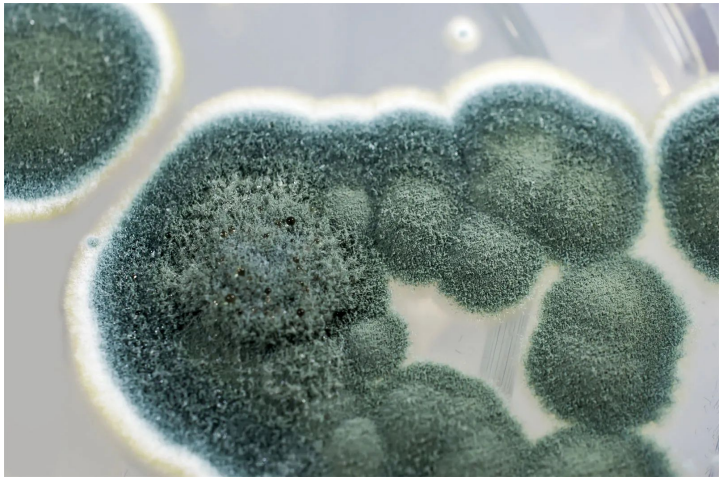
≈ **$1.50**	Defense R&D (Pentagon)
≈ **$1.20**	All civilian science
≈ **65¢**	NIH — all of biomedicine
≈ **12¢**	NSF — all basic science

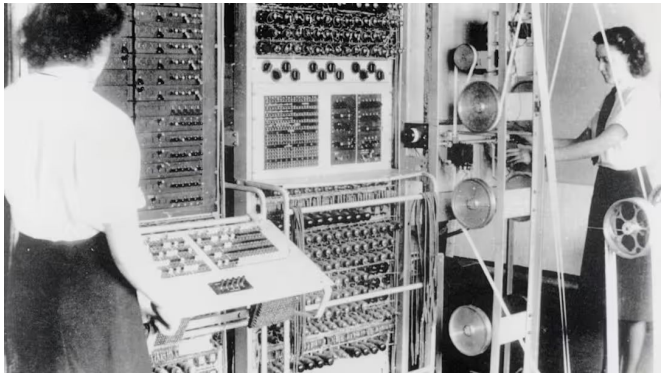
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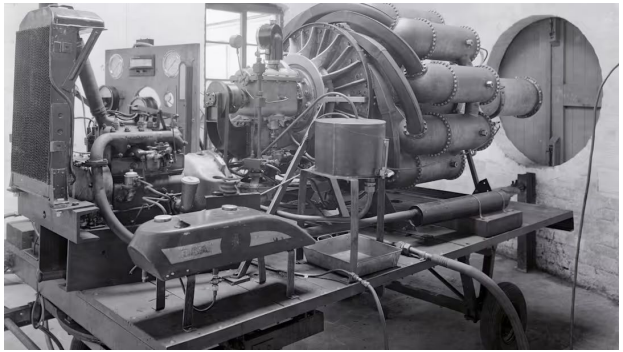
Why does the US Federal Government fund any science?

WWII stimulated some of the greatest scientific advances of 20th Century

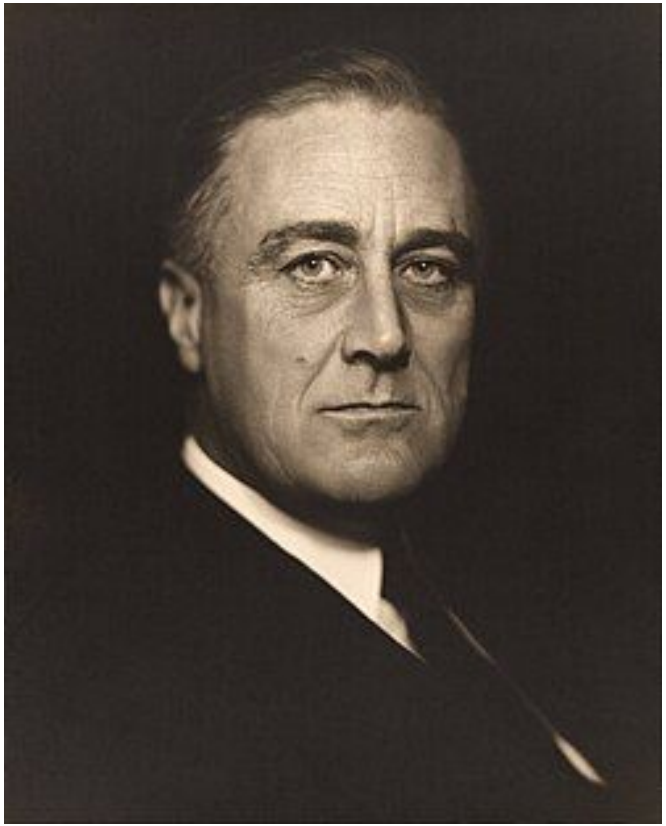




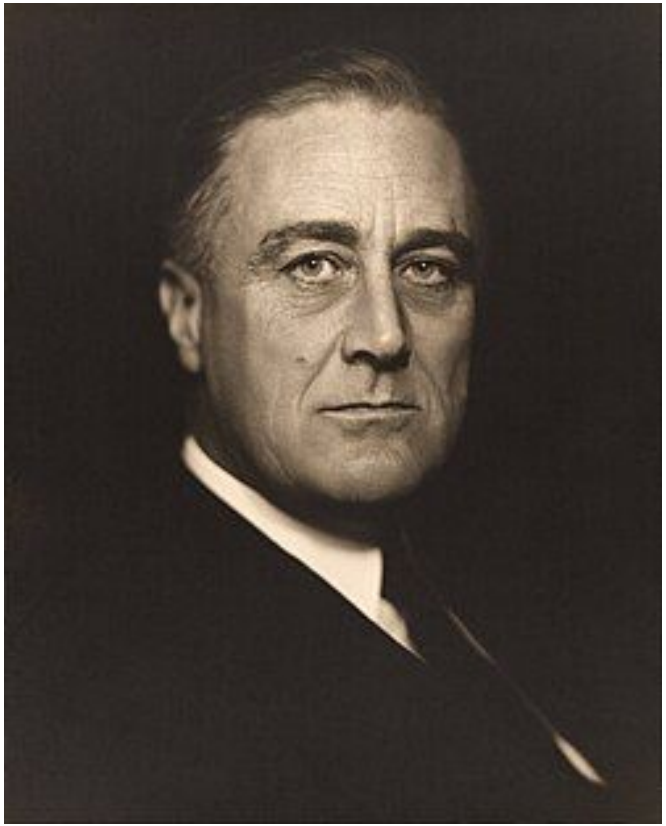












President Franklin D Roosevelt



Vannevar Bush

THE WHITE HOUSE
Washington, D. C.
November 17, 1944

DEAR DR. BUSH: The Office of Scientific Research and Development, of which you are the Director, represents a unique experiment of team-work and cooperation in coordinating scientific research and in applying existing scientific knowledge to the solution of the technical problems paramount in war. Its work has been conducted in the utmost secrecy and carried on without public recognition of any kind: but its tangible results can be found in the communiques coming in from the battlefronts all over the world. Someday the full story of its achievements can be told.

There is, however, no reason why the lessons to be found in this experiment cannot be profitably employed in times of peace. The information, the techniques, and the

SCIENCE

THE ENDLESS FRONTIER

"New frontiers of the mind are before us, and if they are pioneered with the same vision, boldness, and drive with which we have waged this war we can create a fuller and more fruitful employment and a fuller and more fruitful life."

FRANKLIN D. ROOSEVELT.
November 17, 1944.

PART ONE · 1944–1945

Science: The Endless Frontier

ROOSEVELT'S FOUR QUESTIONS · NOV 17, 1944

Release what the war produced

What wartime scientific knowledge can be declassified, so it can start new industries and new jobs?

The war of science against disease

How do we keep the medical research going once the emergency that funded it ends?

What is government's proper role?

How should public and private research relate to one another — and who pays for which?

Where will the next scientists come from?

How do we find and develop scientific talent in American youth, at the scale the war demanded?

Bush's case for federal support

Scientific capital is running out

A nation spends down its stock of basic knowledge as it applies it. America had borrowed most of that capital from Europe — and Europe was in ruins.

Basic research is the pacemaker

"Basic research is performed without thought of practical ends." It is precisely that freedom, Bush argued, that makes it the source of everything practical later.

Industry will not fund it

No company can justify paying for work with no foreseeable product. The payoff is real, distant, and impossible to own — so it falls to government.

The talent pipeline broke

Four years of war pulled a generation out of scientific training. Without deliberate investment in students, there would be no one to do the work.

Vannevar Bush, Science — The Endless Frontier: A Report to the President (July 1945)

WHAT BUSH ACTUALLY PROPOSED · JULY 1945

One New Agency, Five Fundamentals

Bush's answer was institutional, a single new civilian agency, funded by Congress, that would pay for basic research it would never itself perform.

The Five Fundamentals

1. **Stability of funds over years**

   Basic research "ceases to be basic if immediate results are expected on short-term support."

2. **Administered by people who understand research**

   Chosen for their capacity to promote the work, not to represent an interest.

3. **Grants and contracts, never its own laboratories**

   The agency buys research from outside institutions rather than doing it in-house.

4. **Universities keep internal control**

   Policy, personnel, and the scope and method of the research stay with the institution.

5. **Accountable to the President and Congress**

   Independence in the science; democratic accountability for the money.

What actually happened: Congress fought over the plan for five years. The National Science Foundation was created in 1950 — smaller than proposed, with a director appointed by the President rather than the board, and with medicine left to the NIH and defense research to the military. The Foundation Bush designed was never built; the compact he argued for was.

Vannevar Bush, Science — The Endless Frontier (1945), ch. VI "The Means to the End"

1945: The Compact That Built American Science

Government funds curiosity-driven basic research at universities, chosen by scientists through peer review — and the payoff comes later, unpredictably.

What came out of it

The National Science Foundation

Created to fund basic research across science and engineering, chosen by peer review.

Explosive postwar growth of the NIH

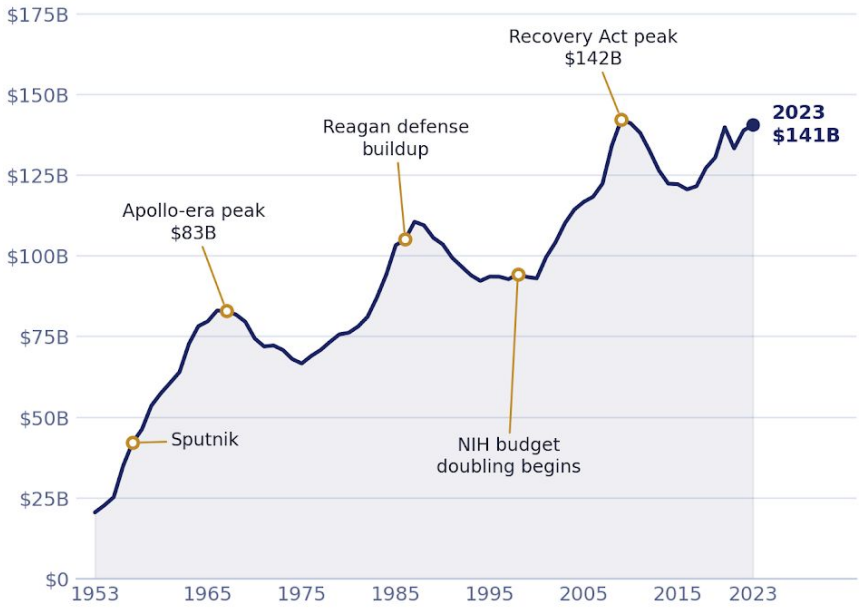
Today the largest funder of biomedical research on Earth — roughly $47B a year.

The “linear model” of innovation

Basic research → applied research → products.

In constant dollars, the money grew

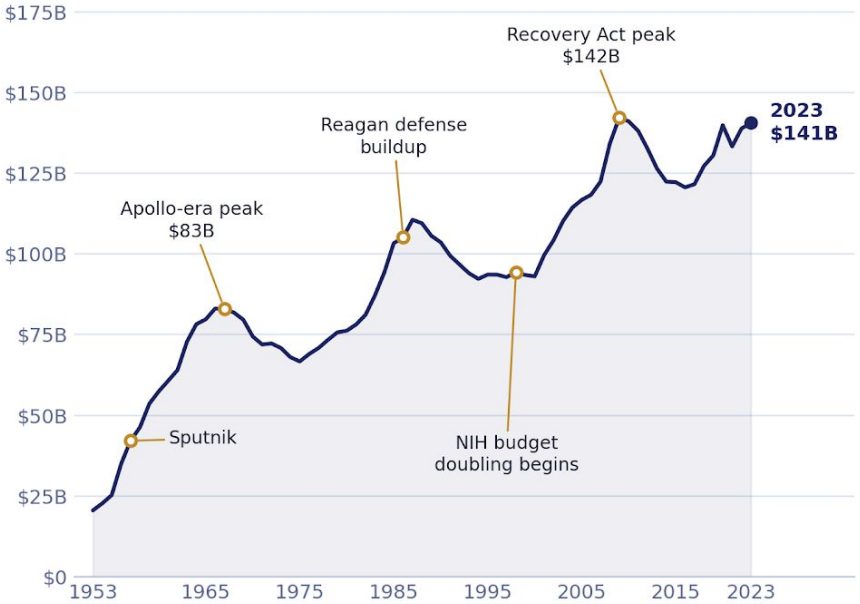
Federal funding of U.S. R&D, billions of constant 2017 dollars



Data: NCSES/NSF, Science & Engineering Indicators, table nsf25326-tab001 (U.S. GDP, R&D, and the ratio of R&D to GDP, 1953–2023).
Constant-dollar series derived by applying the federal share of GDP to real GDP — i.e. deflated by the GDP implicit price deflator, 2017 = 100.

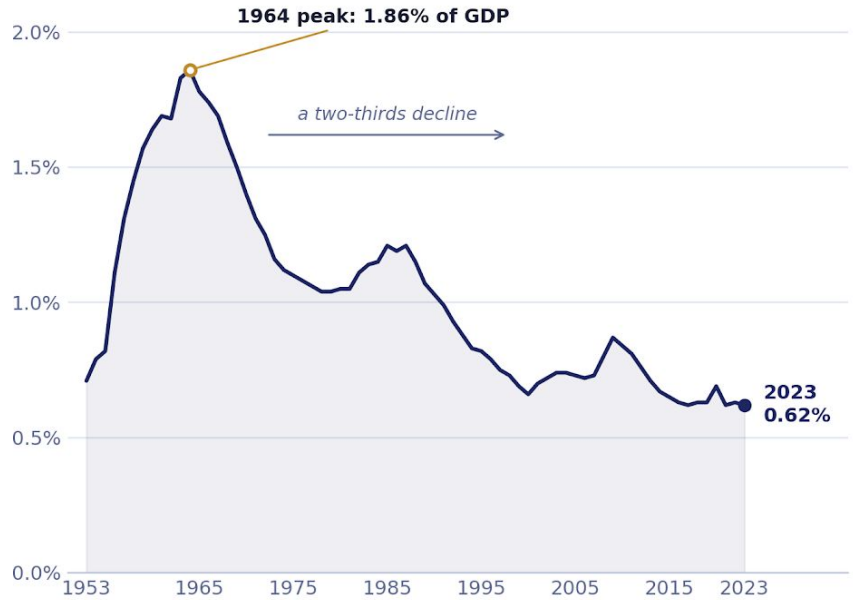
In constant dollars, the money grew

Federal funding of U.S. R&D, billions of constant 2017 dollars



Against the size of the economy, it shrank

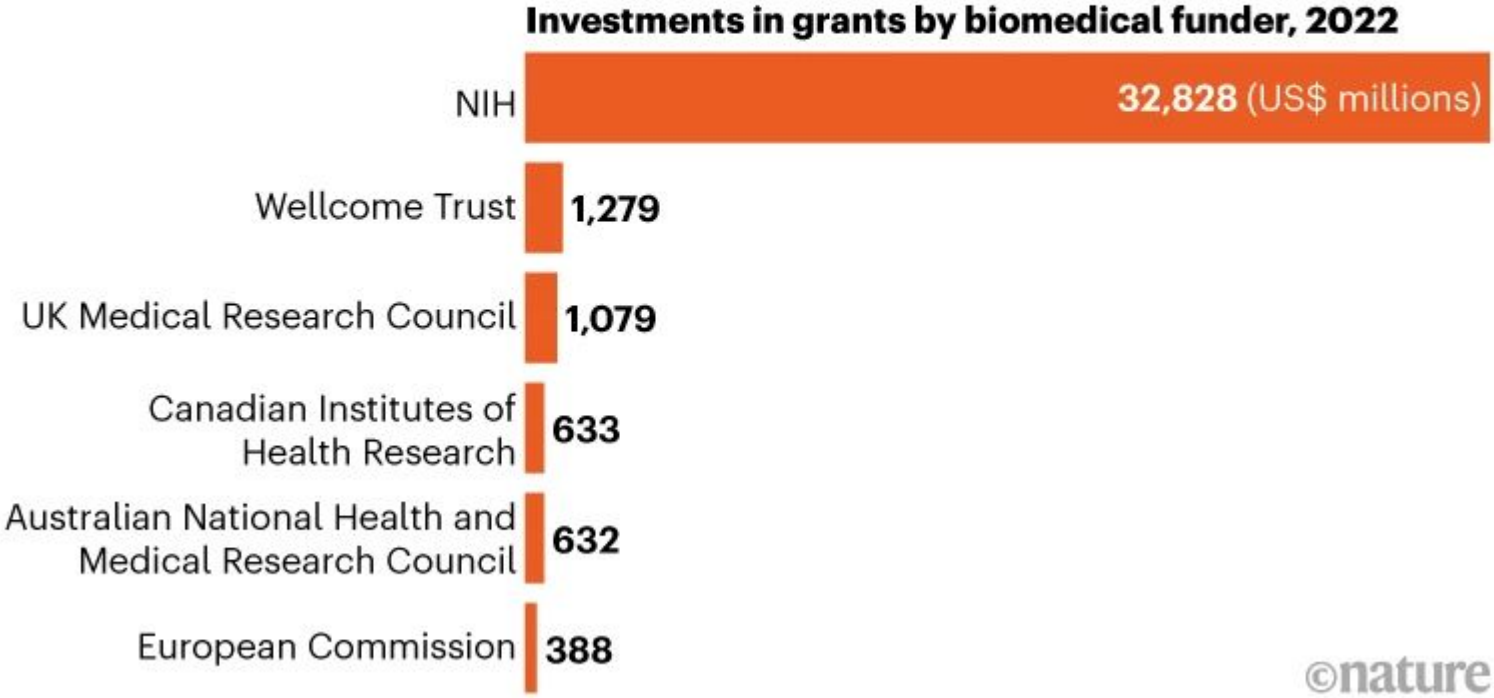
Federally funded R&D as a percentage of GDP



Data: NCSES/NSF, Science & Engineering Indicators, table nsf25326-tab001 (U.S. GDP, R&D, and the ratio of R&D to GDP, 1953–2023).
Constant-dollar series derived by applying the federal share of GDP to real GDP — i.e. deflated by the GDP implicit price deflator, 2017 = 100.

WORLD LEADER

The US National Institutes of Health (NIH) has a budget for biomedical-research grants that towers over the world's next-biggest funders of health research.



How does a scientist access this money: Grants

Write the proposal

~12 dense pages for an NIH R01, plus a mountain of supporting documents.

Study section

A panel of scientists in my field scores it — peer review applied to money. My rivals judge me; I judge them.

>20% funded

Four in five proposals are rejected — many of them excellent.

>1 year

Typical time from idea to funded grant.

>50%+ “overhead”

“Indirect costs” added on top, paid to the university for buildings, compliance, admin.

Politics has often interfered with the idealized vision

APRIL 5, 2009

Fruit flies earn no respect, except among scientists

By Robert S. Boyd

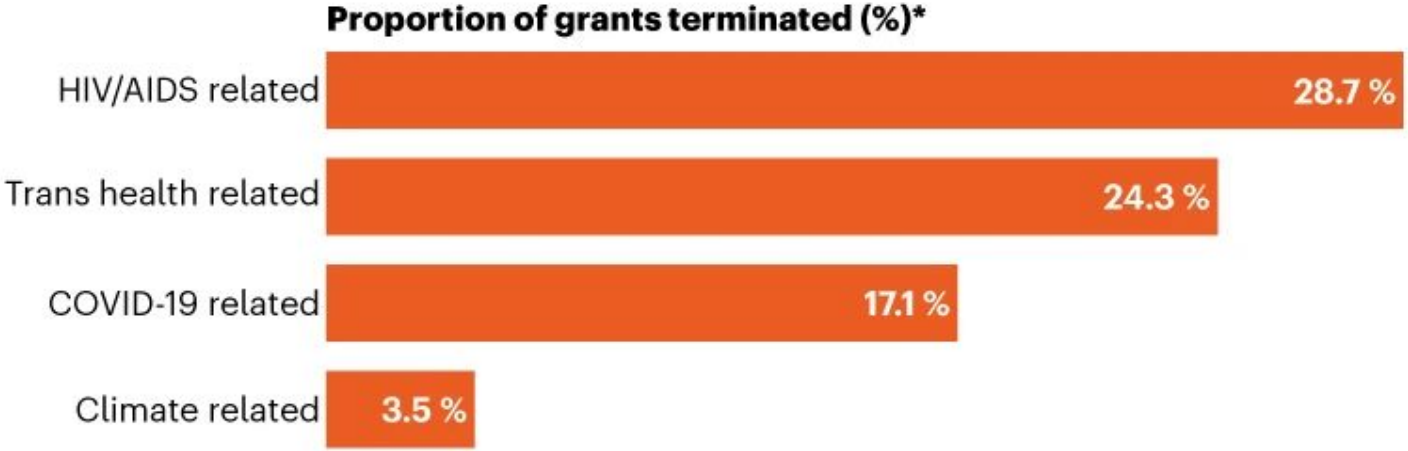
Add as preferred source



That annoying kitchen pest, the fruit fly, occupies an honored place in science and medicine, despite slurs from politicians such as Sen. John McCain and his 2008 sidekick, Sarah Palin.

TERMINATED GRANT TALLY

Under President Donald Trump, the US National Institutes of Health has cancelled roughly 770 active research grants as of 7 April. Nearly 29% of the grants terminated were for research that mentioned HIV/AIDS, and about one-quarter of the grants terminated were related to the health of transgender people. Other topic areas that the Trump team has deemed no longer in the interests of the agency are COVID-19 and the effects of climate change on health.



*Percentages are calculated by searching NIH grant titles and abstracts for key terms such as 'transgender', 'gender diversity', 'climate' and 'SARS-CoV-2'. Percentages do not sum to 100% because these topic do not cover all grants cancelled and because there is some overlap between them.

©nature

SCIENCE
A NEW GOLDEN AGE



A REPORT TO THE PRESIDENT

BY

MICHAEL KRATSIOS
DIRECTOR OF THE WHITE HOUSE OFFICE
OF SCIENCE AND TECHNOLOGY POLICY

JULY 2026

Proposal Rationale: the scientific enterprise is inefficient

1 Productivity has stalled

NIH's budget has more than doubled since the 1990s without a proportional rise in breakthrough treatments.

2 The incumbency tax

Between 1980 and 2008 the average age of an NIH principal investigator rose from 39 to 51; first-time investigators went from their mid-thirties to their forties.

3 Onerous beauracracy

At least 270 new federal requirements were added to research grants after 1991. Some awards take nearly two years. Negotiated indirect cost rates reach 50–60%.

4 Review by consensus

Reviewers scoring the same NIH proposal often reach contradictory conclusions. When one proposal in ten can be funded, panels converge on the least divisive application rather than the most promising.

5 Incentives reward the safe

Tenure rewards volume, so cutting work into smallest-publishable-units beats one ambitious attempt. Negative results, replication, datasets and shared tools earn no credit.

6 No accountability for failure

Of 100 psychology studies someone set out to reproduce, fewer than 40 succeeded. A flawed Alzheimer's paper accumulated 800+ citations and was retracted fifteen years later. No agency pays a price for having rejected a breakthrough.

Michael Kratsios, Science: A New Golden Age — A Report to the President (OSTP, July 2026), ch. II

How to fix science for the new golden age

CHANGE HOW THE CHOICE IS MADE

Golden tickets

Let a single enthusiastic reviewer fund a proposal the full panel would reject.

Fast grants

Decisions in 48 hours on applications that take 30 minutes to write, instead of 6–9 months on applications that take months.

Define Grand Challenges

Moonshot-scale "grand challenges," modeled on the Apollo Program and Human Genome Project

CHANGE WHAT GETS FUNDED

People, not projects

Fund the best scientists over long periods, light reporting, tolerance for early failure. reduce reliance on "legacy institutions" (i.e., universities) and increase investment in industry

Portable fellowships

Expand NSF's Graduate Research Fellowship and let students carry the money to whichever lab they choose.

Emphasize AI, quantum, fusion, space

CHANGE WHO DOES THE SCIENCE

Create X-Labs

Too large for an academic lab, too cross-disciplinary for a single department, and too hard to commercialize for a private corporation

Curiosity-driven institutes

Stable, long-horizon homes for fundamental work, outside the university grant cycle entirely.

STAT+ | POLITICS

White House offers its science blueprint: More AI, less life sciences

'Science: A New Golden Age' report calls for shifting billions from universities to tech companies

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SCIENCEINSIDER | FUNDING

NSF memo implementing Trump's 'golden age' of science unsettles researchers

New document emphasizes support for White House priorities over curiosity-driven ideas

14 SEP 2026 • 3:05 PM ET • BY JEFFREY MERVIS

physicsworld

Topics

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POLICY AND FUNDING | NEWS

Critics deride US administration's 'golden age of science' report

05 Aug 2026 Peter Gwynne

Reading Time: 3 minutes

Assignment:

Find 1 positive change

Find 1 negative change

IT'S NOT ALL FEDERAL

Three Very Different Kinds of Money Fund Science

Federal government

~$200B / yr

Peer-reviewed, risk-averse, concentrated on basic and applied research. The backbone — and the slowest to move.

Industry

~$700B / yr

The majority of all U.S. R&D — but concentrated on development: turning already-known science into sellable products.

Philanthropy

Smaller, faster

Sloan, HHMI, Gates, Simons. Moves fast, takes risks study sections won't, targets neglected problems — but reflects founders' interests.

Industry funds development; the federal government funds the science industry later builds on

What is the return on Federal investment?

What is the return on Federal investment?

NIH, FY2025 (United for Medical Research)

$36.6B

in research grants awarded

390,000+

jobs supported nationwide

$94.2B

in new economic activity generated

$822B

cumulative impact over the last decade

$2.57

generated in economic activity per $1 of NIH grant funding, FY2025

Sources: United for Medical Research (2026)